

## 1. Executive Summary & Application Technical Demands

**Event Land** is an enterprise-grade live ticketing platform built with high concurrency, real-time interactive seating holds, and financial proof verification. Running the system smoothly requires matching the server infrastructure to its core components:

| Stack Component      | Underlying Technology           | Operational Resource Profile on Windows Server                                                                    |
|----------------------|---------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Backend Web API      | .NET 10 (C# Clean Architecture) | Kestrel engine / IIS in-process hosting. Handles REST endpoints, auth, and automated tasks (~350–500 MB RAM).     |
| Real-Time Seat Holds | ASP.NET Core SignalR            | Persistent full-duplex WebSocket connections for live auditorium seat reservation across browsers.                |
| Database Engine      | Microsoft SQL Server (MSSQL)    | Buffer pool caches event catalogs, 4-digit seeded records, and locks. <b>Requires 2.5 GB to 4 GB RAM minimum.</b> |
| In-Memory Cache      | Redis / Memurai Engine          | Provides sub-millisecond atomic seat reservations before committing to the SQL database (~150 MB RAM).            |
| Background Daemon    | PendingBookingExpiryService     | Tick worker running every 60s to expire 30-min unpaid tickets and release seats back to the public pool.          |
| Frontend Client      | React 18 + Vite SPA             | Production build (355 KB gzip) served statically via IIS with instant browser caching.                            |

## 2. Memory Sizing: Why 4 GB RAM Fails & 12 GB RAM Succeeds

**Critical Sizing Warning for Windows Server:** Windows Server with GUI and Remote Desktop (RDP) uses **~2.2 GB RAM idle**. Microsoft SQL Server requires at least **2.0 GB RAM**. On a 4 GB VPS (Micro plan), total baseline memory consumption exceeds 100% immediately on boot, triggering virtual memory disk thrashing, unresponsive RDP, and frequent SQL Server crash out-of-memory errors.

### ❌ MICRO Plan (4 GB RAM) – High Risk

- Windows Server OS: ~2.2 GB
- Microsoft SQL Server: ~2.0 GB
- .NET 10 API & Kestrel: ~0.4 GB
- Total Base Load: ~4.6 GB (Exceeds Physical RAM)**
- Result: Heavy disk paging, laggy RDP, ticket booking crashes.

### ✅ MINI Plan (12 GB RAM) – Recommended

- Windows Server OS: ~2.5 GB
- Microsoft SQL Server: ~4.0 GB (Generous query buffer)
- .NET 10 API + SignalR: ~0.6 GB
- Free Headroom: ~5.0 GB available for burst traffic**
- Result: Blazing fast ticket booking, responsive RDP, zero lag.

### 3. HosterPK Windows VPS Plans & Cost Comparison

Pricing data retrieved directly from [HosterPK Windows VPS](#) and their live billing cart:

| Plan  | vCPU Cores | RAM   | SSD Disk | Bandwidth | Monthly Rate | Annual Cost (1 Year) |
|-------|------------|-------|----------|-----------|--------------|----------------------|
| MICRO | 2 Cores    | 4 GB  | 80 GB    | 8 TB      | Rs. 7,780    | Rs. 93,540 / yr      |
| MINI  | 4 Cores    | 12 GB | 160 GB   | 16 TB     | Rs. 9,450    | Rs. 113,590 / yr     |
| LITE  | 6 Cores    | 20 GB | 280 GB   | 24 TB     | Rs. 14,500   | Rs. 174,000 / yr     |
| PRO   | 10 Cores   | 40 GB | 360 GB   | 28 TB     | Rs. 18,900   | Rs. 226,800 / yr     |

**Why the MINI Plan is the Undisputed Best Investment:** In HosterPK's live cart, the price difference between MICRO (Rs. 93,540/yr) and MINI (Rs. 113,590/yr) is only **Rs. 20,050 for the entire year** (approx. **Rs. 1,670 / month**). In exchange, you get **3x the RAM (12 GB vs 4 GB)**, **2x the CPU Cores (4 vs 2)**, and **2x the SSD Space (160 GB vs 80 GB)**.

## 4. Complete Annual Cost Breakdown (With Existing Domain)

Because you already own your domain for 1 year, you do not need to pay HosterPK for domain registration or transfer. You also leverage Microsoft's free tools to keep total operational expenses at Rs. 0 outside of the server itself:

| Expense Item                      | Service / License Provider                                                | Annual Cost (PKR) |
|-----------------------------------|---------------------------------------------------------------------------|-------------------|
| Cloud VPS Server (1 Year)         | HosterPK MINI Plan (4 Cores, 12 GB RAM, 160 GB SSD, 16 TB BW)             | Rs. 113,590       |
| Domain Name (1 Year)              | Already purchased by you (Existing Domain)                                | Rs. 0 (Included)  |
| Windows Server OS License         | Windows Server 2022 64-bit (Included on HosterPK Windows VPS)             | Rs. 0 (Included)  |
| Database Management System        | Microsoft SQL Server 2022 Express (Official Free Edition, up to 10 GB DB) | Rs. 0 (Free)      |
| Database Administration Studio    | Microsoft SQL Server Management Studio (SSMS)                             | Rs. 0 (Free)      |
| Web Server & Reverse Proxy        | IIS (Internet Information Services) + ASP.NET Core Hosting Bundle         | Rs. 0 (Free)      |
| SSL / HTTPS Security Certificate  | Win-ACME (Automated Let's Encrypt SSL with auto-renewal)                  | Rs. 0 (Free)      |
| TOTAL ESTIMATED ANNUAL INVESTMENT |                                                                           | Rs. 113,590 PKR   |

## 5. Step-by-Step Purchase & Deployment Guide

### Phase 1: Ordering on HosterPK

- 1 Navigate to <https://www.hosterpk.com/windows-vps> and select the **MINI** plan.
- 2 Under Domain Selection, click "**I will use my existing domain and update my nameservers**". Type your registered domain name.
- 3 Select **Windows Server 2022 64-bit** as your Operating System.
- 4 Select Billing Cycle: **Annually (Rs. 113,590)** or **Monthly (Rs. 9,450)**, and complete checkout.

### Phase 2: Connecting Domain & Deploying Stack

- 5 **Map DNS:** In your domain registrar's DNS panel, create two A Records pointing to your new VPS Static IP: @ → VPS\_IP and www → VPS\_IP.
- 6 **Connect via RDP:** On your computer, open **Remote Desktop Connection (mstsc)**, enter your VPS IP, username Administrator, and password.
- 7 **Install Software:** Download and install: (1) .NET 10 Windows Hosting Bundle, (2) SQL Server 2022 Express, (3) SSMS.
- 8 **Deploy Backend & Frontend:** Copy published EventLand.Api and React dist/ folder to C:\inetpub\wwwroot\eventland. Run the API executable once to auto-seed database schema via migration 20260909105956\_InitialCreate.

9 **Install HTTPS:** Run **Win-ACME** command line, select your IIS site, and it automatically binds and auto-renews your Let's Encrypt SSL certificate.